

Учим-ие по таблице

$$(u \cdot v)' = u'v + uv'$$

$$uv = \int u'v dx + \int uv' dx,$$

$$\Rightarrow \int u'v dx = uv - \int uv' dx$$

$$\int u dv = uv - \int v du.$$

$$1) \int \sin ax \cdot x^k dx, \int \cos ax \cdot x^k dx, k \in \mathbb{N}$$

$$2) \int e^{ax} \cdot x^k dx$$

$$\int \sin x \cdot x^2 dx =$$

$$3) \int \sin ax \cdot e^{bx} dx.$$

$$\int \sin x \cdot e^x dx = \sin x e^x - \int e^x \cdot \cos x dx =$$

$$= \sin x e^x - (\cos x e^x + \underbrace{\int \sin x e^x dx}_{I}) =$$

$$I = \frac{\sin x e^x - \cos x e^x}{2} + C.$$

4) Интегралы обратных функций
— по частям.

$$\int \underbrace{1}_{u} \cdot \underbrace{\log_e x}_{v} dx = x \cdot \ln x - \int x \cdot \frac{1}{x} dx =$$

$$= x \ln x - x + C.$$

$$\int \underbrace{1}_{u} \cdot \underbrace{\arctg x}_{v} dx = \arctg x - \int x \frac{1}{1+x^2} dx =$$

$$\int \left(\frac{\ln x}{x} \right)^2 dx = \left[\begin{array}{l} y = \ln x \\ e^y = x \\ dx = x dy \end{array} \right] = \int \frac{y^2}{e^y} dy =$$

$$= \int \underbrace{y^2}_{u} \cdot \underbrace{e^{-y}}_{v'} dy = -y^2 e^{-y} + \int 2y e^{-y} dy =$$

$$= -y^2 e^{-y} - 2y e^{-y} + 2 \int e^{-y} dy = -2 e^{-y} + C.$$

$$\int \frac{x^2}{(1+x^2)^2} dx = -\frac{x}{2(1+x^2)} + \frac{1}{2} \int \frac{1}{1+x^2} dx = \frac{1}{2} \arctg x + C.$$

$$\frac{1}{2} \int \frac{d(1+x^2)}{(1+x^2)^2} = -\frac{1}{2} \cdot \frac{1}{1+x^2}.$$

$$\int (\arcsin x)^2 dx = |\arcsin x = y| =$$

$$\left| \begin{array}{l} x = \sin y; \\ dx = \cos y dy \end{array} \right|$$

$$= \int \underbrace{y^2}_v \underbrace{\cos y}_{u'} dy = \underbrace{y^2 \sin y}_v - 2 \int \underbrace{y}_{u'} \underbrace{\sin y}_{- \cos y} dy =$$

$$= y^2 \sin y - (2y \cdot \cos y + \int \cos y dy) =$$

$$= (\arcsin x)^2 x + 2 \arcsin(\sqrt{1-x^2}) - 2x + C,$$

$$\int \underbrace{\frac{dx}{x^2}}_u \cdot \underbrace{\ln \left| \frac{1-x}{1+x} \right|}_{v'} dx = x^2 \left(2 \left(\frac{1}{x^2-1} \right) \right) - \int \frac{4x}{x^2-1} dx =$$

$$= \frac{2x^2}{x^2-1} - 2 \ln |x^2-1| + C.$$

$$\int x \sin \sqrt{x} dx \left| \begin{array}{l} \sqrt{x} = t \\ x = t^2 \\ dx = 2t dt \end{array} \right| = \int 2t^3 \sin t dt =$$

$$= -2t^3 \cos t + \int 6t^2 \cos t dt = -2t^3 \cos t +$$

$$+ 6t^2 \sin t - \int 12t \sin t dt =$$

$$= -2t^3 \cos t + 6t^2 \sin t - (12t(-\cos t)) +$$

$$+ 12 \int \cos t.$$

$$\int \frac{x^2}{\cos^2 x} dx = x \tan x + \ln |\cos x| + C.$$

Y/3: N1803; 1805; 1807; 1815.

N 1803. $\int \arcsin x dx =$

$$= \int 1 \cdot \arcsin x dx = \left[\begin{array}{l} u = \arcsin x \Rightarrow du = \frac{1}{\sqrt{1-x^2}} \\ dv = dx \Rightarrow v = x + C \end{array} \right] =$$

$$= x \arcsin x + \int x \frac{1}{\sqrt{1-x^2}} dx = \left[\begin{array}{l} u = \frac{1}{\sqrt{1-x^2}}; du = \frac{-1}{1-x^2} \\ dv = x dx \\ v = \frac{x^2}{2} + C \end{array} \right] =$$

$$du = ((\sqrt{1-x^2})^{-1})' = -(\sqrt{1-x^2})^{-2} = -\frac{1}{1-x^2}$$

$$= \frac{x^2}{2\sqrt{1-x^2}} + \int \frac{x^2}{2} \cdot \frac{1}{1-x^2} dx$$

$$\int \frac{x^2 dx}{2(1-x^2)}$$

N 1867

$$\int \frac{x dx}{(x-1)(x+2)(x+3)} =$$

$$\int \frac{x}{(x+1)(x+2)(x+3)} dx$$

$$\frac{x}{(x+1)(x+2)(x+3)} = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{x+3} = \frac{A(x+2)(x+3)}{(x+1)(x+2)(x+3)} + \frac{B(x+1)(x+3)}{(x+1)(x+2)(x+3)} + \frac{C(x+1)(x+2)}{(x+1)(x+2)(x+3)}$$

$$x = A(x+2)(x+3) + B(x+1)(x+3) + C(x+1)(x+2)$$

$$Ax^2 + 5Ax + 6A + Bx^2 + 4Bx + 3B + Cx^2 + 3Cx + 2C$$

$$\begin{cases} A + B + C = 0, \\ 5A + 4B + 3C = 1, \\ 6A + 3B + 2C = 0 \end{cases} \quad \begin{aligned} x &= -\frac{1}{2} \\ y &= 2 \\ z &= -\frac{3}{2} \end{aligned}$$

$$= \int \left(-\frac{1}{2(x+1)} + \frac{2}{x+2} - \frac{3}{2(x+3)} \right) dx =$$

$$= -\frac{1}{2} \int \frac{1}{x+1} dx + 2 \int \frac{dx}{x+2} - \frac{3}{2} \int \frac{dx}{x+3} =$$

$$= -\frac{1}{2} \ln|x+1| + 2 \ln|x+2| - \frac{3}{2} \ln|x+3| + C.$$

$$\#1831 \quad \int \frac{x \ln x}{x^2+1} dx =$$

$$\int (x-1)^2(x+1)^2$$